

Mini Project 1 Data Joining

Libraries

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.3

Import the data files

Each file is in wide format with one row per country and one column per year.

```
sanitation <- read_csv("../data/predictors/at_least_basic_sanitation_overall_access_percent.csv")
fertility <- read_csv("../data/predictors/children_per_woman_total_fertility.csv")
co2 <- read_csv("../data/predictors/co2_pcap_cons.csv")
happiness <- read_csv("../data/predictors/hapiscore_whr.csv")
income <- read_csv("../data/predictors/ilevels4_wb.csv", col_types = cols(.default = "c"))
poverty <- read_csv("../data/predictors/number_of_people_in_poverty.csv")
```

Dropping Year Columns

The year columns from 2023 onwards are dropped first. Each dataset is then changed to long format so that all of them share the same geo, name and year keys, which avoids duplicated year column names when joining.

```
drop_and_pivot <- function(df, value_name) {
  df |>
    select(-any_of(as.character(2023:2100))) |>
    pivot_longer(
      cols = -c(geo, name),
```

```

      names_to = "year",
      values_to = value_name
    ) |>
    mutate(year = as.integer(year))
  }

sanitation_long <- drop_and_pivot(sanitation, "sanitation_access_pct")
fertility_long <- drop_and_pivot(fertility, "children_per_woman")
co2_long <- drop_and_pivot(co2, "co2_per_capita")
happiness_long <- drop_and_pivot(happiness, "happiness_score")
income_long <- drop_and_pivot(income, "income_level")
poverty_long <- drop_and_pivot(poverty, "people_in_poverty")

```

Join into a single dataset

```

combined <- list(
  sanitation_long, fertility_long, co2_long,
  happiness_long, income_long, poverty_long
) |>
  reduce(full_join, by = c("geo", "name", "year")) |>
  arrange(geo, year)

glimpse(combined)

```

Rows: 43,485

Columns: 9

```

$ geo           <chr> "afg", "afg", "afg", "afg", "afg", "afg", "afg", ~
$ name          <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Af~
$ year          <int> 1800, 1801, 1802, 1803, 1804, 1805, 1806, 1807, ~
$ sanitation_access_pct <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ children_per_woman <dbl> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, ~
$ co2_per_capita <dbl> 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, ~
$ happiness_score <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ income_level   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ people_in_poverty <dbl> 3082964, 3082964, 3082964, 3082964, 3082964, 308~

```

```
range(combined$year)
```

```
[1] 1800 2022
```

```
n_distinct(combined$geo)
```

```
[1] 195
```

```
head(combined)
```

```
# A tibble: 6 x 9
```

	geo	name	year	sanitation_access_pct	children_per_woman	co2_per_capita
	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	afg	Afghanist~	1800	NA	7	0.001
2	afg	Afghanist~	1801	NA	7	0.001
3	afg	Afghanist~	1802	NA	7	0.001
4	afg	Afghanist~	1803	NA	7	0.001
5	afg	Afghanist~	1804	NA	7	0.001
6	afg	Afghanist~	1805	NA	7	0.001

```
# i 3 more variables: happiness_score <dbl>, income_level <chr>,
```

```
#   people_in_poverty <dbl>
```